Summary by Group

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Introduction

Split-apply-combine is the workhorse pattern of applied data analysis: split the data by a grouping variable, apply a summary function to each group, and combine the results. It produces the per-group means, medians, counts, and other statistics that populate Table 1 of most scientific papers.

Prerequisites

The reader should know what the mean, median, and SD are, and be comfortable with R's `dplyr` pipe syntax.

Theory

For a dataset with n observations and a grouping variable g taking values $g_1, \ldots, g_k$, split-apply-combine computes

$$s_j = f(\{x_i : g_i = g_j\})$$

for each group j, where f is the summary function (mean, median, quantile, any aggregator). The returned object has k rows, one per group.

R implements this via base `aggregate()`, `tapply()`, and `by()`, and via the modern `dplyr::group_by() |> summarise()` and `data.table` equivalents.

Assumptions

Purely descriptive; no distributional assumption. The grouping variable should have enough observations per group for the summary to be meaningful (a mean based on $n = 2$ is noisy).

R Implementation

```
library(dplyr)
library(data.table)
library(gtsummary)

set.seed(2026)
trial <- tibble::tibble(
  arm      = factor(rep(c("Placebo", "Active"), each = 75)),
  sex      = factor(sample(c("F", "M"), 150, replace = TRUE)),
  age      = round(rnorm(150, 58, 12)),
  hba1c    = round(c(rnorm(75, 7.3, 0.7), rnorm(75, 6.9, 0.8)), 1),
  event    = rbinom(150, 1, 0.15)
)

# dplyr
trial |>
```

```r
  group_by(arm) |>
  summarise(
    n          = n(),
    age_mean   = mean(age),
    age_sd     = sd(age),
    hba1c_med = median(hba1c),
    hba1c_iqr = IQR(hba1c),
    event_rate = mean(event)
  )

# Multi-variable grouping
trial |>
  group_by(arm, sex) |>
  summarise(across(c(age, hba1c), list(mean = mean, sd = sd)),
            .groups = "drop")

# data.table
dt <- as.data.table(trial)
dt[, .(n = .N, age_mean = mean(age), hba1c_median = median(hba1c)),
   by = .(arm)]

# Publication-ready
trial |>
  tbl_summary(by = arm, include = c(age, sex, hba1c, event),
              statistic = list(all_continuous() ~ "{mean} ({sd})"))
```

Output & Results

```
# A tibble: 2 x 6
  arm         n age_mean age_sd hba1c_med hba1c_iqr event_rate
  <fct> <int>    <dbl>  <dbl>     <dbl>     <dbl>      <dbl>
1 Placebo  75     57.9   12.1       7.3       0.9      0.173
2 Active   75     57.8   11.8       6.9       1.1      0.120
```

The two arms differ in HbA1c (medians 7.3 vs. 6.9) while being comparable in age and sex distribution – the typical pattern in a Table 1 balance check.

Interpretation

Report grouped summaries for every important subgrouping in a study: treatment arms, sex, study site, risk stratum. The `gtsummary::tbl_summary(by = ...)` call produces a fully formatted Table 1 with p-values on request.

Practical Tips

- Always include `n` per group so readers know the denominator.
- Handle missing values explicitly: `summarise(mean(x, na.rm = TRUE))` or pre-filter.
- For many summary statistics per variable, `across(everything(), list(mean = mean, sd = sd))` is concise.
- `data.table` is faster for very large datasets; the syntax differs but the logic is identical.
- Report median + IQR for skewed variables and mean + SD for approximately normal ones; mix the two when the Table 1 covers both types.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1